

Unit 1 Review*Equations and Inequalities* **ANSWER KEY**

HOW TO USE THIS. This review covers every learning target from all ten lessons of Unit 1 — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Equivalent Forms of Rational Numbers*Lessons 1 and 2 · MA.7.NSO.1.2 · 8 questions*

1. Write $\frac{7}{8}$ as a decimal and as a percent.

 *$7 \div 8 = 0.875$, then multiply by 100.***Answer.** *0.875 and 87.5%*

2. Write $\frac{5}{9}$ as a decimal and as a percent.

 *$5 \div 9 = 0.5555\ldots$ so the bar goes over the 5.**A calculator shows 0.55555556 — the last digit is rounded, not exact.***Answer.** *$0.\overline{5}$ and $55.\overline{5}\%$*

3. Write $0.\overline{45}$ as a fraction in simplest form.

*Two repeating digits, so the denominator is 99: $45/99 = 5/11$.***Answer.** *$5/11$*

4. Write 0.375 as a fraction in simplest form.

 *$375/1000$, divided top and bottom by 125.***Answer.** *$3/8$*

5. Write 128% as a decimal and as a fraction.

 *$128 \div 100 = 1.28$, and $128/100$ reduces by 4.***Answer.** *1.28 and $32/25$ (or $1\frac{7}{25}$)*

6. Order from least to greatest: $0.\overline{6}$, $\frac{5}{8}$, 63%, 0.605

*Write them all as decimals: 0.605, 0.625, 0.63, 0.666...***Answer.** *$0.605 < \frac{5}{8} < 63\% < 0.\overline{6}$*

7. A calculator shows 0.55555556 when you enter $5 \div 9$. What is the exact value, and why does the display end in a 6?

Answer. *The exact value is $0.\overline{5}$. The calculator has a ten-digit display, so it rounds the last digit it can show.*

8. Which is greater, $\frac{4}{7}$ or 0.57? Show how you know.

$4 \div 7 = 0.5714\dots$, and $0.5714\dots > 0.57$.

Answer. *$\frac{4}{7}$ is greater.*

PART B. Solving Two-Step Equations

Lessons 3 and 5 · MA.7.AR.2.2 · 9 questions

9. **SSDD.** A phone plan costs \$32 to start, plus \$9 a month.

The four parts look alike. They are not the same question — that is the whole item, and it is why the label is on this page and not on theirs.

- a. Write an expression for the total cost after m months.

No total is given, so there is nothing to solve and no equal sign.

Answer. *$32 + 9m$*

- b. The total came to \$158. Write and solve an equation for m .

$9m = 126$

$126 \div 9 = 14$, and $32 + 9(14) = 158$ ✓

Answer. *$32 + 9m = 158$, so $m = 14$ months*

- c. Write and solve an inequality for the whole number of months that keeps the total at \$200 or less.

$9m \leq 168$, so $m \leq 18.66\dots$

Round DOWN: 18 months costs $32 + 162 = \$194$ ✓, and 19 months costs \$203 ✗

Answer. *$32 + 9m \leq 200$, so $m \leq 18.66\dots$, and the answer is 18 months*

- d. What does the 32 stand for, and what does the 9 stand for?

No arithmetic. This part is asking whether the letters mean anything to you.

Answer. *32 is the one-time starting cost. 9 is the cost of each month.*

Every part starts from the same two numbers. Part a is an expression and has no equal sign; part b is an equation; part c is an inequality whose answer must be rounded DOWN because it is a ceiling; part d asks what the numbers mean and needs no arithmetic at all.

10. $5x - 8 = 37$

$5x = 45$

Answer. *$x = 9$*

11. $-3n + 11 = -16$

$-3n = -27$

Answer. $n = 9$

12. $y/4 + 6 = 1$

$y/4 = -5$

Answer. $y = -20$

13. $7(x - 3) = 42$

$x - 3 = 6$ (divide both sides by 7)

Answer. $x = 9$

14. $-2(m + 4.5) = 15$

$m + 4.5 = -7.5$ (divide both sides by -2)

Answer. $m = -12$

15. $\frac{2}{3}(p - 6) = 8$

$p - 6 = 12$ (multiply both sides by $3/2$)

Answer. $p = 18$

16. $0.4(t + 25) = 6$

$t + 25 = 15$ (divide both sides by 0.4)

Answer. $t = -10$

17. Solve $3(x + 5) = 27$ two ways. Show both.

Distribute: $3x + 15 = 27$, so $3x = 12$ and $x = 4$.

Divide first: $x + 5 = 9$, so $x = 4$.

Answer. $x = 4$

PART C. Writing and Solving Equations from a Situation

Lessons 4, 6 and 7 · MA.7.AR.2.2 · 6 questions

18. A gym charges \$45 to join plus \$22 a month. A member has paid \$221 in all. How many months of membership is that?

$m = \text{the number of months. } 45 + 22m = 221$

$22m = 176$, so $m = 8$

Answer. **8 months**

19. A restaurant bill was reduced by a \$9 coupon and then split evenly among 6 friends. Each friend paid \$12.50. What was the bill before the coupon?

$b = \text{the bill before the coupon. } (b - 9) / 6 = 12.50$

$b - 9 = 75, \text{ so } b = 84$

Answer. \$84.00

20. Five concert tickets were bought and a \$3.25 service fee was added to each one. The total was \$91.25. What was the price of one ticket before the fee?

$t = \text{the price of one ticket. } 5(t + 3.25) = 91.25$

$t + 3.25 = 18.25, \text{ so } t = 15$

Answer. \$15.00

21. Elena earns \$520 a week plus a 4% commission on her sales. Last week her total earnings were \$732. How much did she sell?

$s = \text{her sales. } 520 + 0.04s = 732$

$0.04s = 212, \text{ so } s = 5,300$

Her commission was \$212 — not \$732.

Answer. \$5,300

22. A coat is marked 35% off. After the discount, Kai uses a \$12 coupon and pays \$53.00. What was the original price?

$p = \text{the original price. } 0.65p - 12 = 53$

$0.65p = 65, \text{ so } p = 100$

35% off means paying 65%.

Answer. \$100.00

23. A tank held 3,600 gallons and is draining at 145 gallons per hour. It now holds 1,570 gallons. How long has it been draining?

$h = \text{the number of hours. } 3,600 - 145h = 1,570$

$145h = 2,030, \text{ so } h = 14$

Answer. 14 hours

PART D. Solutions to Inequalities, and Graphing Them

Lesson 8 · MA.8.AR.2.2 · 6 questions

24. Circle every value that is a solution to $2x - 7 \geq 5$. -2 0 5.9 6 6.1 12

$2x \geq 12, \text{ so } x \geq 6.$

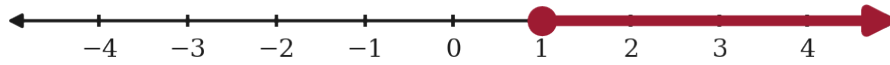
6 counts because the symbol has the equal bar. 5.9 does not.

Answer. 6, 6.1, 12

25. Solve and graph: $4x - 3 \geq 1$

$4x \geq 4$

Answer. $x \geq 1$



26. Solve and graph: $-2x + 3 > 13$

$-2x > 10$, then divide by -2 and REVERSE.

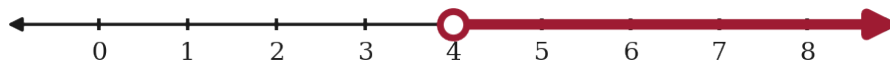
Answer. $x < -5$



27. Solve and graph: $x/2 - 1 > 1$

$x/2 > 2$, then multiply both sides by 2.

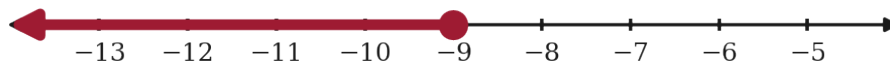
Answer. $x > 4$



28. Solve and graph: $7 - x \geq 16$

$-x \geq 9$, then multiply by -1 and REVERSE.

Answer. $x \leq -9$



29. Explain how you know that 0.5 is not a solution to $3x + 1 > 2.5$.

Answer. Substituting gives $3(0.5) + 1 = 2.5$, and $2.5 > 2.5$ is false. The two sides are equal, and the symbol is strict.

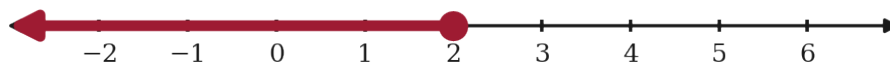
PART E. Inequalities with the Distributive Property

Lesson 9 · MA.8.AR.2.2 · 5 questions

30. Solve and graph: $4(x + 1) \leq 12$

$x + 1 \leq 3$ (divide both sides by 4 — no reversal)

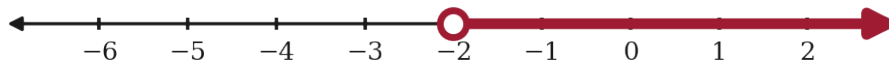
Answer. $x \leq 2$



31. Solve and graph: $-3(x - 1) < 9$

$x - 1 > -3$ (divide both sides by -3 and REVERSE)

Answer. $x > -2$



32. Solve and graph: $\frac{1}{5}(x + 6) < 3$

$x + 6 < 15$ (multiply both sides by 5 — no reversal)

Answer. $x < 9$



33. Solve $-2(3x - 5) \geq 22$ two ways. Show both.

Distribute: $-6x + 10 \geq 22$, so $-6x \geq 12$ and $x \leq -2$ (reversed at the division).

Divide first: $3x - 5 \leq -11$ (reversed here), so $3x \leq -6$ and $x \leq -2$.

Answer. $x \leq -2$

34. Sam solved $-4(x + 2) > 8$. He distributed the -4 and reversed the symbol, then divided by -4 and reversed again. What went wrong, and what is the correct solution?

Distributing rewrites one side. It is not an operation on both sides, so nothing reverses there.

Sam reversed twice, which returns the symbol to where it started, so he got $x > -4$ — backwards.

Correct: $-4x - 8 > 8$, so $-4x > 16$, and $x < -4$ (one reversal, at the division).

Answer. The correct solution is $x < -4$.

PART F. Inequalities in Real-World Situations

Lesson 10 · MA.8.AR.2.2 · 5 questions

35. Write an inequality: a ride requires riders to be at least 48 inches tall. Use h for a rider's height in inches.

“At least” includes the number, so the symbol is $\geq$ and the graph would use a closed circle.

Answer. $h \geq 48$

36. A field trip costs \$180 plus \$14 for each student. The budget is \$630. How many students can go? Justify your answer.

s = the number of students. $180 + 14s \leq 630$

$14s \leq 450$, so $s \leq 32.14...$

Round DOWN — it is a maximum. 32 students cost \$628; 33 would cost \$642.

Answer. 32 students

37. Marisol needs at least \$185 for a bike. She has \$40 and will save the same amount each week for 8 weeks. What is the least she can save each week?

$w = \text{the amount saved each week. } 40 + 8w \geq 185$

$8w \geq 145, \text{ so } w \geq 18.125$

Round UP — it is a minimum. At \$18.12 she reaches \$184.96 and misses; at \$18.13 she reaches \$185.04.

Answer. *\$18.13 a week*

38. A store sells $\frac{3}{4}$ of the items on a shelf, then brings out 60 more. The store keeps at least 200 items on the shelf. How many items were on the shelf to begin with?

$x = \text{the number of items to begin with. One quarter are left, so } \frac{1}{4}x + 60 \geq 200.$

$\frac{1}{4}x \geq 140, \text{ so } x \geq 560$

Answer. *At least 560 items*

39. A phone plan costs at most \$50 a month. It charges \$28 plus \$0.11 for each gigabyte of data. How many whole gigabytes can be used?

$g = \text{the gigabytes used. } 28 + 0.11g \leq 50$

$0.11g \leq 22, \text{ so } g \leq 200$

This one is already a whole number, so nothing needs rounding.

Answer. *200 gigabytes*

PART G. Looking Ahead: the Unknown on Both Sides

Later this year · MA.8.AR.2.1 · 2 questions

40. Solve $5(n - 2) = 3n + 8$.

$5n - 10 = 3n + 8$ (distribute first)

$2n = 18$

Answer. *$n = 9$*

41. Plan A costs \$60 to join plus \$8 an hour. Plan B costs \$20 to join plus \$12 an hour. At how many hours do the two plans cost the same?

$60 + 8h = 20 + 12h$

$40 = 4h, \text{ so } h = 10$

Answer. *10 hours, at \$140 each*